***73939133** is an amazing prime number. You can chop any number of digits off the end and still end up with a prime. It's the largest known prime with this property.*

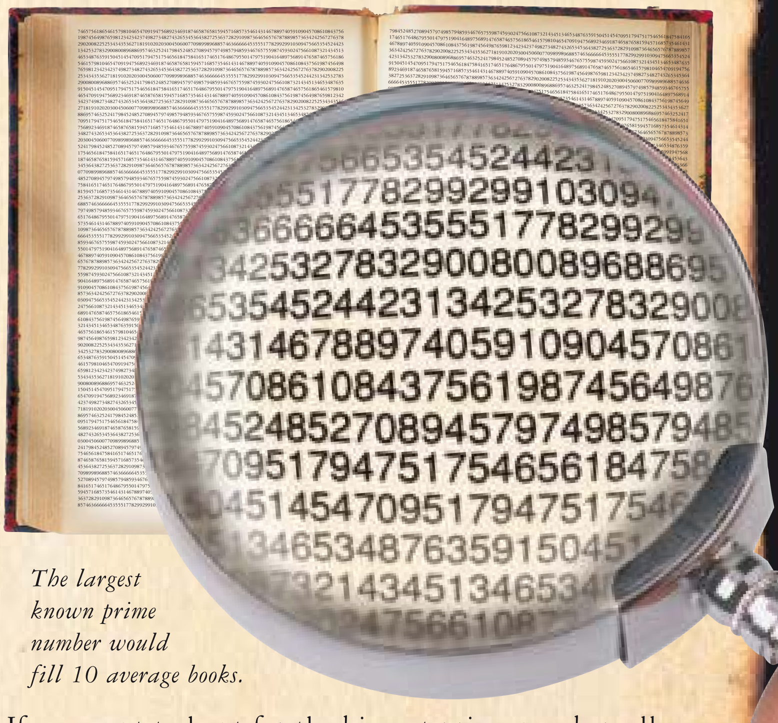
The hunt for the biggest primes

A sieve is handy for finding small primes, but what about big ones? Is 523,367,890,103 a prime number? The only way to be sure is to check nothing will divide into it, and that takes time. Even so, mathematicians have found some amazingly big prime numbers. The biggest so far is more than 7.8 million digits long. If you tried to write it in longhand, it would take 7 weeks to write and would stretch for 46 km (29 miles).

$100,000

REWARD

for the first person to find a prime number with more than ten million digits



The largest known prime number would fill 10 average books.

If you want to hunt for the biggest prime number, all you need do is download a program from the web and let your computer do the rest. Worldwide, 40,000 people are doing exactly this. The first person to find a prime with more than 10 million digits will win a $100,000 prize.

29 31 37 41 43

Secret codes

Multiplying primes together is easy, but what about doing the reverse – splitting a number into its prime "factors"? For really big numbers, this is virtually impossible. In fact it's so difficult, it makes prime numbers perfect for creating unbreakable secret codes. When you spend money on the internet, your details are hidden by a code made this way. The "lock" for the code is a huge number, and the "key" consists of the number's prime factors.

Prize numbers

Secret codes made from prime numbers are so reliable that one company in the USA has offered a prize to anyone who can crack their code. If you can find the two prime numbers that multiply to give the number below, you'll win $20,000. Here's the number:

310741824049004372135075003588856793003734602284272754572016194882320644051808150455634682967172328678243791627283803341547107310850191954852900733772482278352574238645401469173660247765234660 9.

Prime timing

Some insects use prime numbers for protection. Periodical cicadas spend exactly 13 or 17 years underground as larvae, sucking roots. Then they turn into adults, swarm out of the ground, and mate. Both 13 and 17 are prime numbers, so they can't be divided into smaller numbers. As a result, parasites or predators with a life-cycle of, say, two or three years, almost never coincide with a swarm.